

ANALYTIC SOLUTION OF THE GENERAL ANHARMONIC-OSCILLATOR PROBLEM

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ABSTRACT. The regular and irregular solution of the differential three-dimensional Schrödinger equation with the confining potential $V(r) = g_{-2}r^{-2} + g_{-1}r^{-1} + g_1r + \dots + g_{2q}r^{2q}$, $q \geq 1$, is presented. In full analogy with the recent $q = 1$ conjecture of Singh *et al.* [1], the corresponding Green's function is constructed in terms of the extended continued fraction. For the simplest example with $V(r) = r^2 + \alpha^2 r^4$, its convergence is proved and the resulting energy spectrum is numerically tested.

Recently, the radial Schrödinger equation

$$-\frac{d^2}{dr^2}\psi(r) + \frac{l(l+1)}{r^2}\psi(r) + V_{(q)}(r)\psi(r) = E\psi(r) \quad (1)$$

with the polynomial confining potential

$$V_{(q)}(r) = \frac{g_{-2}}{r^2} + \frac{g_{-1}}{r} + \sum_{j=1}^{2q} g_j r^j, \quad g_{2q} = \alpha^2 > 0, \quad g_{-2} > -\frac{1}{4} - l(l+1) \quad (2)$$

was solved by Singh *et al.* [1] for $q = 1$. Their wave functions may be given the compact form [2]

$$\psi(r) = r^\sigma \exp[-f(r)]\phi(r), \quad \sigma = \sigma_\pm = \frac{1}{2} \pm \left(\frac{1}{2} + g_{-2} + l^2 + l\right)^{1/2},$$

$$f(r) = \sum_{j=1}^{q+1} \beta_j r^j / j, \quad \beta_{q+1} = g_{2q}^{1/2} = \alpha > 0, \quad (3)$$

$$\beta_{q-j} = \frac{1}{2\alpha} \left(g_{2q-j-1} - \sum_{i=1}^j \beta_{q+1-i} \beta_{q+i-j} \right), \quad j = 0, 1, \dots, q-1,$$

$$\phi(r) = \sum_{n=0}^{\infty} d_{n+1} r^n$$

which generalizes directly the well-known special-function representation of the Coulombic ($q = 0$) or harmonic-oscillator ($g_{-1} = g_1 = 0$) eigenstates [3].

It is easy to verify that even for an arbitrary $q \geq 1$, the Taylor coefficients may be given by the same algebraic formula

$$d_{n+1} = (-1)^n \frac{\det Q(n)}{b_1 b_2 \dots b_n} d_1, \quad n = 1, 2, \dots$$

$$Q(n) = \begin{pmatrix} a_1 & b_1 & 0 & \dots & 0 \\ c_2^{(1)} & a_2 & b_2 & 0 \dots & 0 \\ \dots & & & & \\ c_{q+1}^{(q)} & c_{q+1}^{(q-1)} & \dots & & 0 \\ \dots & & & & \\ 0 & \dots & 0 & c_n^{(q)} \dots c_n^{(1)} & a_n \end{pmatrix} \quad (4)$$

$$b_n = n(n + 2\sigma - 1), \quad a_n = -g_{-1} - 2\beta_1(n + \sigma - 1),$$

$$c_n^{(1)} = E + \beta_1^2 - \beta_2(2n + 2\sigma - 3),$$

$$c_n^{(j)} = -g_{j-1} + \sum_{i=0}^{j-1} \beta_{i+1} \beta_{j-i} - \beta_{j+1}(2n + 2\sigma - j - 2),$$

$$j = 2, 3, \dots \min(q, n - 1), \quad n = 1, 2, \dots$$

Indeed, this formula represents the unique solution of the recurrence relations

$$Q(\infty) \begin{pmatrix} d_1 \\ d_2 \\ \dots \end{pmatrix} = 0 \quad (5)$$

which result from the insertion of the power-series ansatz (3) into the Schrödinger Equation (1). In this way, we have proved

PROPOSITION 1. *For an arbitrary integer $q \geq 0$, the algebraic formulae (3) and (4) represent the general solution of the differential Equation (1).*

The choice of the sign σ_+ or σ_- corresponds to the solution $\psi_{\pm}(r)$ which is regular or irregular in the origin [3], respectively. The $r \rightarrow \infty$ boundary conditions (normalizability requirements) are to be imposed on the physical states $\psi_{\pm}(r)$ in addition. In accord with the standard oscillation theorems [4], this may be used in the numerical specification of the energy levels $E = E_0$.

The analytic approach characterizing E_0 as poles of the Green's function was initiated in [1] for $q = 1$. It is a matter of straightforward algebra to generalize this construction to $q > 1$ as well.

The main idea [5] is to decompose $Q(\infty)$ in (5) into the product of the three triangular matrices, $Q(\infty) = -Q_{(u)}Q_{(d)}Q_{(1)}$. The technical details may also be found in Reference [5] and imply

PROPOSITION 2. *The Green's function of the Schrödinger Equation (1) may formally be represented by the extended continued fraction (ECF) quantify $f_1^{(\infty)}(E)$ defined by the recurrences*

$$-\frac{1}{f_k^{(N)}} = a_k + b_k f_{k+1}^{(N)} c_{k+1}^{(1)} + \dots + b_k f_{k+1}^{(N)} b_{k+1} f_{k+2}^{(N)} \dots b_{k+q-1} f_{k+q}^{(N)} c_{k+q}^{(q)},$$

$$f_{N+1}^{(N)} = 0, \quad k = N, N-1, \dots, 2, 1$$
(6)

in the limit $N \rightarrow \infty$.

It is important to note here that even for the simplest Singh's $q = 1$ case, the uniqueness and the rigorous foundation of the $N \rightarrow \infty$ limiting transition is, in fact, not entirely trivial – in detail, this will be discussed elsewhere [2]. In what follows, the most important new features of the present $q > 1$ ECF construction will be illustrated in more detail on the particular $q = 2$ example.

PROPOSITION 3. *For the s-wave (quartic) anharmonic oscillator potential $V_A = r^2 + \alpha^2 r^4$ the ECF representation of the Green's function is convergent.*

Proof. The finite sequence $F_{N-m} = (N-m)f_{N-m}^{(N)}(E)/\alpha$, $m = 0, 1, \dots, N-1$ satisfying Equation (6) may be divided into three separate groups:

(1) For the first few m 's, we get the explicit form of terms $F_{N+3-3k} = k^{-1}(1 + 0(1/N))$, $F_{N+2-3k} = 2(k+1-2\epsilon)(1 + 0(1/N))$, $\epsilon = \alpha^2 E + \frac{1}{4}$, $F_{N+1-3k} = k(k+1-2\epsilon)/(2\rho_N^3)(1 + 0(1/N))$, $\rho_n^3 = 2\alpha^4(n+1)^2$, which exhibits the $q+1$ -periodic oscillations ($q = 2$). In the region $1 - 2\epsilon < k < k_1 = \rho_N \sqrt[3]{2} \gg 1$, all F 's are positive and the smallest term F_{N+1-3k} grows to reach the next value of $F_{N+3-3(k+1)}$ for $k = k_1$.

(2) In the next part of the asymptotic region $k_1 < k < k_2 \ll N$, we put $k = k_1 + t$, $t = 0, 1, \dots$. From (6) we get the less pronounced oscillatory behaviour of $F_{N+3-3t} = A_t/k_1$, $F_{N+2-3t} = 2B_t/k_1$, $F_{N+1-3t} = C_t/k_1$, $N_1 = N - 3k_1$, since A_t, B_t and C_t are equal to $1 + 0(1/k_1)$. Inclusion of the next corrections gives $A_t = C_t = (1 + 1/k_1)^{-t}$, $B_t = (1 + 2/k_1)^{-t}$ and implies the relative decrease of F_{N+2-3t} towards the level of the remaining two values. Roughly speaking, we may choose $k_2 = k_1(1 + 1/n^2)$ and get, in the leading order, the smooth dependence of F_{N_2-m} on $m = 0, 1, \dots, m \ll N_2 = N - 3k_2$ since $(1 + 1/k_1)^{k_1} = 2.71 \dots$ for $k_1 \gg 1$.

(3) Defining $F_n^{(0)} = (1 - \epsilon F_n^{(0)} + \rho_n^3 F_n^{(0)^2})^{-1}$ for $n < N_2$, we may subtract the dominant stationary contribution $F_n^{(0)} = 1/\rho_n - \frac{1}{3}\rho_n^2 + \dots$ from the values F_n . If $n \gg 1$, the smallness of the new quantities $R_n = F_n - F_n^{(0)}$ enables us to linearize the corresponding modification $(R_{n+1}, R_{n+2}) \rightarrow R_n$ of the mapping (6) and the rest of proof of the ECF convergence may proceed in close analogy with the classical continued fraction theory [6]. The relative error of the ECF quantities F_k generated from the 'fixed-point' initialization $F_{N_2}^{(0)}$, $N_2 \gg k$, may be made arbitrarily small by the choice of the 'cutoff' $N_2 \gg 1$. This completes the proof.

We may summarize:

(a) The slow rate of convergence following from the preceding proof was observed even in Reference [1] ($q = 1$) and the numerical applications of the formalism were not recommended there. Although an increase of q worsens this situation even more, the decisive and highly efficient acceleration of convergence is obtained by using the fixed-point initialization (cf. [5]) and the averaging over the surviving $q + 1$ -periodic oscillations (cf. also [7]), in practice. Thus, for the lowest energy levels at least, the present technique compares favourably with the specialized numerical approaches. For $V = V_A$, their small sample is discussed in Reference [8] — our Table I

TABLE I
Convergence of energies obtained for the anharmonic oscillator potential $V_{A0}(r) = r^2 + r^4$ from the ECF secular equation.

N^a	root E_{00}	N^a	root E_{01}	N^a	root E_{02}
21	4.843	201	13.54	1001 ⁺	23.178
22	4.447	202	14.25	1002 ⁺	23.266
23	4.695	203	12.57	1003 ⁺	23.473
201	4.6476	501 ⁺	13.1585	2001 ⁺	23.2951
202	4.6523	502 ⁺	13.1615	2002 ⁺	23.2949
203	4.6466	503 ⁺	13.1506	2003 ⁺	23.3022
(21-3)	4.659 ^b	(201-3)	13.432 ^b	(1001-3 ⁺)	23.3057 ^b
(201-3)	4.64884 ^b	(201-3 ⁺)	13.1615 ^b	(2001-3 ⁺)	23.29744 ^b
		(501-3 ⁺)	13.15685 ^b		

^a ECF initialization $F_N = 1/\rho_N$. The improvement $F_N = 1/\rho_N - 1/3\rho_N^2$ is marked by crosses.

^b Extrapolated energy mean value averaged over the three neighboring N 's.

demonstrates that the anharmonic oscillator energies become consistent with the results of Reference [9] for the reasonably small cutoffs N . The underlying algorithm is almost trivial — we have used the small programmable desk-calculator (Compucorp 327).

(b) The main merit of our general $q \geq 1$ solution is not only the direct analogy of $\psi(r)$ with the harmonic-oscillator states but first of all the possibility to extend the analytic methods of Reference [1] to $q > 1$ via the new ECF concept. In a way, the ECF representation of the Green's function is one of the most natural 're-summations' of the divergent perturbation series. In our particular example, secular equation $1/f_1^{(\infty)}(E_0) = 0$ displays clearly the well-known non-perturbative character of the anharmonic oscillator energy. Similar singularities appear also in the Taylor coefficients ($d_2 = d_1/2\alpha$) and exponential factor ($\beta_1 = 1/2\alpha$) of the exact, convergent and normalizable wave function $\psi(r)$.

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